

Adversarial Sample Based on Structured Fusion Noise for Botnet Detection in Industrial Control Systems

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Abstract—The industrial control system’s artificial intelligence-based botnet intrusion detection system has a high detection performance and efficiency in an environment without interference. However, these systems are not immune to evasion through adversarial samples. In this study, we introduce a feature extraction technique tailored for ICS botnet detection. This approach classifies traffic packets based on network traffic attributes and ICS-specific identification codes, encompassing the statuses of ICS devices, enhancing detection precision. Meanwhile, this strategy addresses challenges in ICS data collection and bolsters experimental efficacy. To build a comprehensive botnet intrusion dataset within an ICS, we concurrently utilized existing ICS devices to collect both standard ICS and botnet traffic. Additionally, we present an innovative adversarial sample generation method for botnet detection models, integrating both time-domain and frequency-domain noise. Testing under three real-world ICS attack scenarios revealed our technique can markedly degrade the classification performance of eight leading AI-based detection models, emphasizing its potential for evading AI-based ICS intrusion detectors.

Index Terms—Adversarial sample, botnet, industrial control system, fusion noise.

I. INTRODUCTION

IN DISTINGUISHING between the industrial control system and the traditional internet, a key difference lies in the integration of ICS-specific protocols with standard network traffic in the former. The volume of industrial control traffic far surpasses that of standard internet traffic, owing to the frequent

interactions of ICS equipment. Botnets produce significant traffic, comprising both Denial of Service (DoS) attack packets, the number of traffic packets is roughly the same as the industrial control protocol traffic. Notably, their protocol type, payload size, and packet transmission intervals are distinct. While the sophistication in botnet detection is advancing, so is the capability to evade these detection systems. Many models, despite their enhanced accuracy and efficiency in detection and optimized hardware, have overlooked robustness, rendering them vulnerable to savvy attackers. Within the network security and intrusion detection domain, attackers often employ adversarial samples to deceive AI-based detection models and bypass systems. The techniques for generating adversarial samples in network traffic intrusion detection can be categorized into gradient-based attacks, score optimization-based attacks, Generative Adversarial Network (GAN)-based attacks, and alternative model-based attacks.

Adversarial samples can introduce different forms of noise perturbations based on the method employed for their creation. However, the overarching objective remains the same: to inject noise into the original sample. A subset of random signal processes, termed as ‘colored noise’ or ‘environmental noise,’ represents time series. One of its notable types, Gaussian noise, aligns with the probability density function of time domain sampling and adheres to a normal distribution. Naturally occurring noises, such as blue, pink, red, and purple noise, have power spectral density functions in the frequency domain that follow a $1/f^\alpha$ distribution, where α indicates the rate of frequency decay.

Noise has versatile applications across domains such as transfer learning, data augmentation, adversarial sample generation, and adversarial training. In our research, we generate adversarial samples leveraging colored noise whose frequency-domain sampled power spectral density function aligns with the $1/f^\alpha$ distribution and Gaussian noise, which corresponds to the time-domain sampled probability density function of a normal distribution. Our innovative approach, merging both time-domain and frequency-domain sampled noise for adversarial sample creation, aims to bolster the model’s resilience and enhance defenses against adversarial attacks. Below are our key contributions:

(1) To precisely extract the traffic and industrial control attributes of the botnet, we introduce a feature extraction method that amalgamates the address details from standard network protocols with the distinctive identification codes of

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industrial control protocols for industrial control protocol and botnet protocol in ICS. This delineation between industrial control traffic and botnet traffic markedly enhances performance. Guided by this approach, we collected millions of industrial control traffic packets and botnet packets. At the same time, we've curated an ICS botnet intrusion dataset enriched with both traffic and industrial control characteristics. This dataset can be used directly for AI-based model training. Authentic industrial control traffic and botnet traffic were collected within a real-world industrial control system setting.

(2) We propose an adversarial sample generation technique for black-box attacks using fusion noise. This method combines colored noise derived from frequency-domain sampling with a power spectral density function adhering to the $1/f^\alpha$ distribution and Gaussian noise from time-domain sampling that conforms to a normal distribution. The resultant adversarial samples, integrating both noise types, markedly diminish the model's detection capabilities.

(3) In an industrial control setting, our adversarial sample generation technique demonstrates strong performance. According to empirical tests, it can diminish the detection capabilities of four leading neural network models and four advanced machine learning intrusion detection models. Given the unique nature of industrial control system, we initially deploy our proposed adversarial sample technique within this context.

II. RELATED WORKS

In this section, we review the existing literature on intrusion detection in ICS, focusing on detection models, feature extraction and data processing methods, and adversarial sample attacks.

A. Intrusion Detection Models

Researchers have developed a variety of advanced model architectures to enhance detection accuracy and efficiency. In the Internet of Things (IoT) domain, which is closely related to ICS, Kim et al. [1] proposed a hybrid model that integrates machine learning and deep learning techniques, offering the flexibility to classify multiple types of botnet attacks. To capture the temporal dependencies in botnet traffic, Imrana et al. [2] introduced a Bidirectional Long Short-Term Memory (BiLSTM) network, which significantly improved detection performance through bidirectional data processing.

As model complexity increased, researchers began exploring ensemble and hybrid architectures. Ali et al. [3] designed the ACLR model, which stacks ANN, CNN, LSTM, and RNN layers to effectively capture complex attack patterns and features of botnets. Wazzan et al. [4] proposed a Cross CNN-LSTM framework for IoT botnet detection, mapping the MITRE ATT&CK model into the detection process and introducing the kill-chain concept to enable early warning.

In addition, several studies have focused on optimizing specific models. Chen et al. [5] extracted features using CNNs and incorporated Particle Swarm Optimization (PSO) to enhance the traditional SVM algorithm. Mudassir et al. [6] developed a multi-layer deep learning model for industrial

IoT botnet classification, achieving an accuracy of up to 99%. Palla and Tayeb [7] proposed a deep learning-based method to detect Mirai malware targeting IoT nodes, addressing the overfitting problem by carefully tuning the number of hidden layers and neurons. Liao et al. [8] leveraged a combination of CNN and Transformer architectures to jointly learn the spatial and temporal characteristics of IoT traffic, thereby improving the model's ability to recognize minority attack classes. Laghrissi et al. [9] constructed an intrusion detection system (IDS) based on LSTM and proposed three related LSTM-based variants collectively referred to as PCA +MI.

B. Feature Extraction and Data Processing

In industrial control systems (ICS), effective feature processing is a crucial means of improving intrusion detection performance, as it enables models to focus on the most informative and discriminative attributes. High-quality features and balanced datasets form the cornerstone of model performance [10]. In terms of feature extraction, Alqahtani et al. [11] employed the Fisher score to identify key features and eliminate irrelevant ones, optimizing classification performance by minimizing intra-class distance and maximizing inter-class distance. Abbasi et al. [12] utilized a linear low-cost algorithm to assign feature weights, effectively mitigating the impact of redundant or irrelevant attributes.

In ICS-specific contexts, data processing plays a particularly critical role. Jiang and Chen [13] analyzed traffic from industrial protocols such as Modbus, TCP, and S7, employing a Denoising Autoencoder (DAE) for feature extraction and combining SMOTE and Tomek-Link mechanisms to address the problem of class imbalance. Similarly, Mengara et al. [14] proposed innovative solutions to cope with class imbalance and high-dimensional data in intrusion detection. Meidan et al. [15] developed a network-based anomaly detection approach for IoT systems by extracting statistical features that capture behavioral snapshots of benign IoT traffic, demonstrating potential advantages in binary classification.

Other researchers have focused on dataset-specific processing algorithms. Rohmantlyo et al. [16] proposed customized data processing, training, and testing algorithms for the N-BaIoT dataset. Gupta and Singh [17] designed a multi-layer detection framework incorporating the SHAP method to explain model predictions, thereby improving system transparency and interpretability. Alqahtani and Aldhyani [18] developed an IoT botnet detection application using a hybrid deep learning model trained on the N-BaIoT dataset, which was generated from nine commercial IoT devices by injecting BASHLITE and Mirai attacks. Liu et al. [19] introduced a lightweight IoT intrusion detection approach based on correlation coefficients and variance inflation factors (VIF), performing flow-level feature selection to enhance efficiency. Zong et al. [20] proposed an automated fuzzing method for industrial control protocols (ICPs) based on the Denoising Diffusion Probabilistic Model (DDPM), which improved fuzzing effectiveness, identified protocol vulnerabilities, and reduced potential economic risks and impacts.

C. Adversarial Attacks

Despite continuous advancements in detection technologies, the robustness of models has become an increasingly prominent issue, as attackers frequently exploit adversarial samples to evade detection. When attackers possess knowledge of a model's internal parameters and architecture, they can launch gradient-based or optimization-based attacks [21]. Mohammadian et al. [22] employed the Jacobian Saliency Map Algorithm (JSMA) to determine optimal feature combinations, enabling attacks with minimal feature perturbations. Haroon and Ali [23] applied classical gradient-based attacks such as FGSM and JSMA to several cybersecurity datasets to evaluate the defensive performance of machine learning algorithms.

In cases where model internals are inaccessible, black-box attacks are commonly adopted. Liu et al. [24] integrated genetic algorithms into an IoT detection framework, proposing a black-box testing system (TLTD) for adversarial evaluation. Wu et al. [25] introduced a deep reinforcement learning-based black-box attack, capable of generating adversarial traffic to deceive detection models effectively. Liu et al. [26] developed a network traffic obfuscation technique to resist traffic classification, proposing an improved gradient-based attack method adapted to practical strategies and conducting extensive black-box experiments across different model combinations. Similarly, Wang et al. [27] proposed a projection-based adversarial attack framework (ProGen), which significantly reduced the detection performance of multiple machine learning models.

Usama et al. [28] leveraged Generative Adversarial Networks (GANs) to manipulate specific classes within data-sets and craft adversarial samples. To address the gradient vanishing problem in traditional DCGANs, Alghazzawi et al. [29] introduced a hybrid optimization algorithm, where the generated adversarial samples caused the true positive rate (TPR) of an SVM-based detection model to plummet from 80% to 0%.

In response to such attacks, researchers have proposed corresponding defense mechanisms. Ding et al. [30] employed adversarial training as an active defense strategy to enhance model robustness by retraining on adversarial samples, and further incorporated a denoising autoencoder as a passive defense approach.

III. ADVERSARIAL SAMPLE GENERATION METHOD

The current AI-driven botnet intrusion detection system for industrial control system focuses solely on detecting anomalies in clean samples, neglecting the system's robustness in identifying perturbations in adversarial samples. Our study addresses this gap.

We employed a two-pronged noise generation approach: in the time domain, we generated Gaussian noise that aligns with a normal distribution, and in the frequency domain, we produced colored noise using power spectral density functions that adhere to a $1/f^\alpha$ distribution. By fusing these noises, adversarial samples were crafted to test the model's resilience.

Our experimental setup involved simulating botnet attacks on industrial control equipment. Traffic from these regular

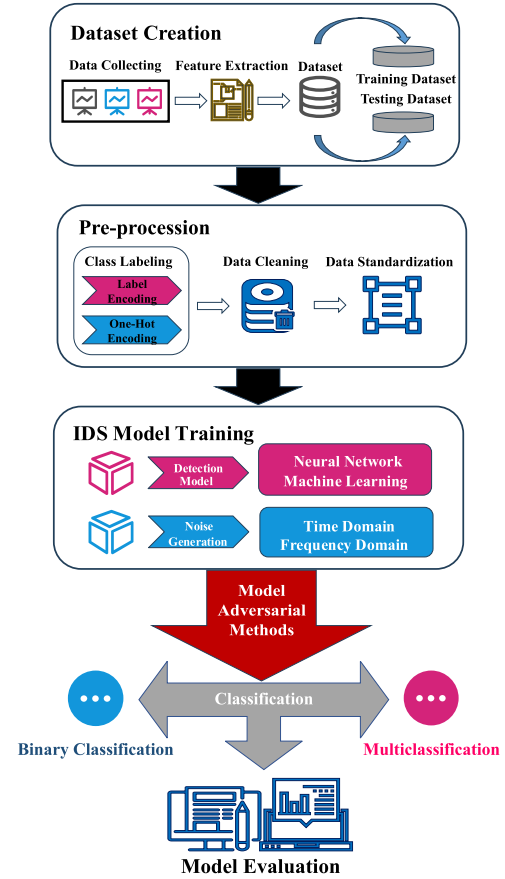


Fig. 1. Adversarial sample generation framework.

operations was first captured, and pertinent features were extracted from the traffic packets. The ensuing data underwent a series of preprocessing steps, which included categorizing labels, data cleansing, and normalization. An adversarial sample creation model was then trained independently of the detection model. This newly generated adversarial sample, influenced by the combined noise, was then introduced to the detection model.

To evaluate the model's robustness against these adversarial intrusions, we relied on a confusion matrix based on the model's output. Performance metrics such as Accuracy, Precision, Recall, and F1-Score were subsequently derived. The corresponding process is illustrated in **Fig. 1**.

A. Data Collection

We established a control mode featuring a Siemens master with two slave configurations. The slaves connect to the master through distinct channels: one via a serial port and the other through a network port. The Siemens PLC, leveraging the S7 protocol, managed functionalities such as temperature adjustments, motor speed controls, tri-color light switches, and other analog and digital automation controls. This PLC communicates with the HMI seamlessly. For meter regulation tasks, the Schneider controller interfaced with the HMI using the Modbus-TCP protocol. From the operator's perspective, the HMI provides real-time monitoring of various parameters of the industrial control equipment. The architecture of this

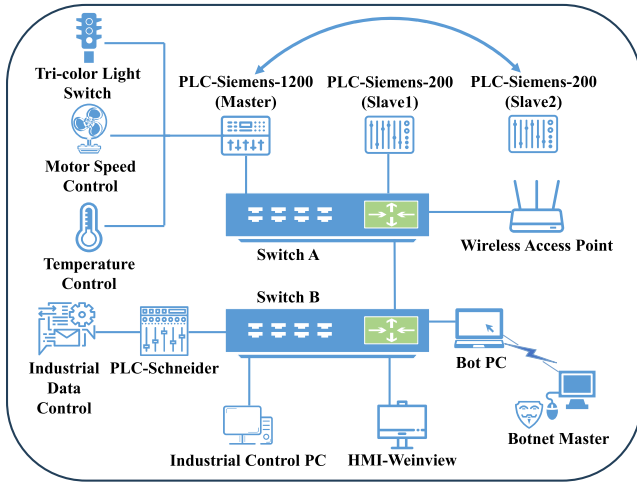


Fig. 2. Construction industrial control environment.

comprehensive industrial control environment is illustrated in Fig. 2.

The HMI-Weinview and PLC-Schneider interact using the Modbus-TCP protocol, whereas the PLC-Schneider and the meter communicate using the Modbus-RTU protocol. We set up a botnet to control a Bot PC and used three DoS command attacks, namely the SYN Flood Attack, UDP Flood Attack, and TCP Garbage Attack, to perform DoS attacks on the PLC-Schneider device's port 502 and prevent it from normally communicating with the HMI-Weinview and preventing operators from operating it normally. Additionally, we gathered traffic packets from both the industrial control device's routine operations and its attacks. We collected millions of pieces of data, of which the normal traffic is made up of about 800,000 packets and the anomalous traffic is made up of roughly 200,000 packets. The normal traffic also includes industrial control data transmitted using the S7 protocol and the Modbus protocol.

B. Feature Extraction

We propose a 9-tuple feature extraction method. Let's look at the source IP address, destination IP address, source MAC address, destination MAC address, communication protocol, source port, and destination port of the conventional network protocol $C = \{c_1, c_2, \dots, c_7\}$. Industrial control protocol unique identifier $I = \{i_1, i_2\}$, representing modbus_trans_id and modbus_func_code in Modbus-TCP protocol, the above features form a 9-tuple.

The conventional protocol 7-tuple is an extension of the traditional protocol 5-tuple by including the MAC address. While the 5-tuple can uniquely identify traditional network protocols and extract inter-flow information based on it [32], integrating the MAC address offers an added layer of defense against certain attacks, notably the man-in-the-middle attacks.

In the context of industrial control system, the 2-tuple modbus_trans_id differentiates standard network traffic from traffic specific to the industrial control protocol. Furthermore, it aids in extracting distinct communication attributes. Using the modbus_func_code, Modbus-TCP industrial control traffic

can be segmented based on specific functionalities, which subsequently allows for the identification of the characteristics of the various commands sent to the industrial control devices.

For a more nuanced extraction of DoS traffic characteristics, both uplink and downlink traffic are considered simultaneously. The 9-tuple claims that it is possible to extract the traffic features of both conventional network protocols and industrial control protocols. Additionally, it suggests the potential to glean industrial control data directly from Modbus packets, providing a means to deter potential unauthorized modifications to the control data by adversaries. We combine network traffic characteristics and industrial control device characteristics to improve detection accuracy.

Ultimately, from this process, we derive two sets of features: the traffic feature represented as $X = \{x_1, x_2, \dots, x_p\}$ and the industrial control feature denoted as $Y = \{y_1, y_2, \dots, y_q\}$.

We extracted a dataset containing 27 features using 9-tuples. The total number of traffic features is 21, including network uplink traffic and downlink traffic. These include the number of traffic packets, load size, and flow duration, respectively. Also included are the mean, maximum, minimum, and standard deviation of packet loads and transmission intervals. The operational features of 6 devices are included in the industrial control features. These include electrical energy, voltage, current, active power, power factor and frequency of industrial control equipment.

Unlike Internet traffic, ICS communication exhibits high periodicity, strict timing control, and low variance under normal conditions. While the extracted 21 traffic-related features (e.g., mean, standard deviation, maximum, and minimum packet sizes or time intervals) may appear conventional, their inclusion is specifically motivated by the temporal and statistical behavior of ICS traffic. Therefore, even small deviations in these statistical indicators can effectively reflect abnormal behaviors caused by botnet-induced DoS activities. The combination of first-order (mean) and second-order (variance-related) statistics enables the model to capture both steady-state operational characteristics and dynamic fluctuations in ICS data flows. Table I and Table II display the flow features and industrial control features respectively.

C. Dataset Classification

Within industrial control system, standard industrial control traffic significantly surpasses intrusion attack traffic. However, DoS attack traffic, unlike Man-in-the-Middle (MITM), Backdoor, and Injection attacks, carries an extensive amount of data. For more effective training outcomes in our experiment, we sifted through millions of packets to find some of the traffic packets, and we curated a subset of data for evaluation. This encompassed 10,000 samples each of normal traffic, SYN Flood attacks, UDP Flood attacks, and TCP Garbage attacks, aggregating to 40,000 samples.

The data distribution among the sets was as follows: the training set comprised 70% (28,000 samples), the validation set 10% (4,000 samples), and the test set 20% (8,000 samples). Importantly, each traffic type was proportionally represented within each set, ensuring consistency. Furthermore, every

TABLE I
FLOW FEATURES

Name	Value	Name	Value	Name	Value
up_pkt_count	count of uplink packets	down_payload_std	standard deviation of downlink packet payload	down_pkt_count	count of downlink packets
up_time_interval_mean	mean of uplink packet time interval	up_payload_count	payload of uplink packets	down_time_interval_mean	mean of downlink packet time interval
down_payload_count	payload of downlink packets	up_time_interval_min	minimum of uplink packet time interval	up_payload_mean	mean of uplink packet payload
down_time_interval_min	minimum of uplink packet time interval	down_payload_mean	mean of downlink packet payload	up_time_interval_max	maximum of uplink packet time interval
up_payload_min	minimum of uplink packet payload	down_time_interval_max	maximum of downlink packet time interval	down_payload_min	minimum of downlink packet payload
up_time_interval_std	standard deviation of uplink packet time interval	up_payload_max	maximum of uplink packet payload	down_time_interval_std	standard deviation of downlink packet time interval
down_payload_max	maximum of downlink packet payload	duration	network flow duration	up_payload_std	standard deviation of uplink packet payload

TABLE II
INDUSTRIAL CONTROL FEATURES

Name	Value	Name	Value
total_electrical_energy	equipment electrical energy	active_power	equipment active power
voltage	equipment voltage	power_factor	equipment power factor
current	equipment current	frequency	equipment frequency

model in our experiments was trained and evaluated using these identical sets.

D. Data Preprocessing

The category labels of the data we propose to capture contain benign, SYN, UDP, TCP, a total of four different categorical values. Because the machine learning model cannot directly take these values, the strings must first be encoded before being input into the model for training and testing. Therefore, we encode the category labels of the model with one-hot encoding and label encoding.

In this study, the attack traffic consists of TCP and UDP packets originating from a Bot DoS attack targeting the PLC-Schneider. Notably, these packets lack industrial control data, necessitating the processing of missing values in this segment. On the other hand, the normal traffic packets represent interactions between industrial control devices utilizing the Modbus-TCP protocol, and they inherently contain industrial control data. Given that the industrial control data captured by the device can only be altered manually through a button press, and considering that the botnet DoS attack doesn't modify the device's operational value, this specific data is incorporated into the features extracted from the attack traffic for our analysis.

To ensure authenticity and maintain operational normality, a minor fluctuation is introduced to the industrial control data. This ensures that it remains within acceptable operational limits. Subsequently, these adjusted data features are appended to our dataset. Concurrently, we implemented data standardization to normalize the dataset's features. The primary objective of this process is to level the playing field among features by removing any inherent disparities in their value ranges. A dataset with disparate ranges across its features can severely compromise the performance of a classification model. By standardizing the data, not only do we simplify the training process, but we also enhance the computational efficiency of the model.

Standardization is an essential preprocessing step, given the disparate scales of the extracted features. We utilized z-score standardization, which centers each feature at a mean of 0 and scales it to a standard deviation of 1. This conversion to a consistent scale is critical for optimizing the training of neural network architectures and other scale-sensitive algorithms, fostering stable convergence and improving model performance.

E. Adversarial Sample Generation

We propose an attack method for botnet intrusion detection models that fuses noise from both the time and frequency

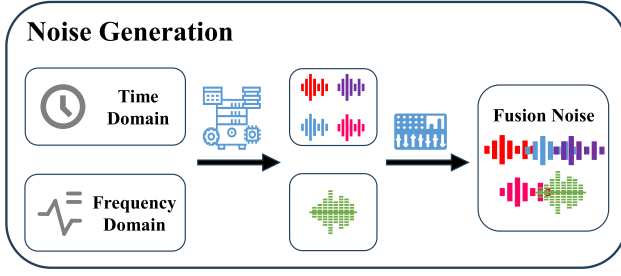


Fig. 3. Fusion noise generation.

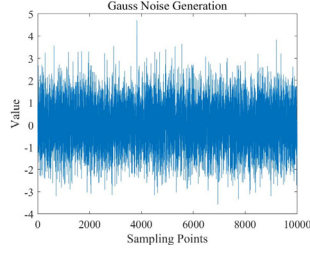


Fig. 4. Gaussian noise generation.

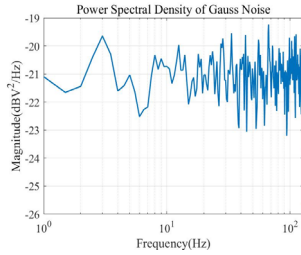


Fig. 5. Gaussian noise power spectral density.

domains. This hybrid perturbation combines two components: time-domain Gaussian noise (which conforms to a normal distribution) and frequency-domain colored noise (which satisfies the $1/f^\alpha$ distribution). Our adversarial sample generation involves first synthesizing five distinct noise components: Gaussian noise and four colored noises. Following this, all components are standardized to a mean of 0 and a variance of 1. Finally, these standardized components are combined to create the fusion noise used for the adversarial attack.

The fusion noise generation process is shown in **Fig. 3**.

Gaussian noise is generated by python library function, which conforms to the standard normal distribution with mean 0 and standard deviation 1 distribution, and its probability density function satisfies:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (1)$$

where μ stands for the mean and σ stands for the standard deviation. In order to correspond to 10,000 pieces of data for each category of the dataset, we generated 10,000 Gaussian noises conforming to the standard normal distribution.

The result of Gaussian noise generation is shown in **Fig. 4**, **Fig. 5**, **Fig. 6** and **Fig. 7**.

The power spectral density function of colored noise in the frequency domain satisfies the $1/f^\alpha$ distribution, and

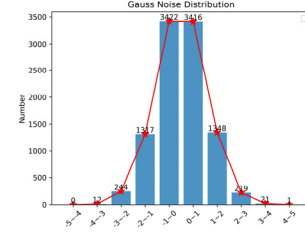


Fig. 6. Gaussian noise distribution.

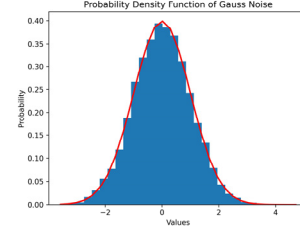


Fig. 7. Gaussian noise probability density function.

TABLE III
CHARACTERISTICS OF COLORED NOISE

Noise Type	α	PSD Slope Dependence	Spectral Slope Rate
Red Noise	2	$\propto 1/f^2$	-6dB/oct -20dB/dec
Pink Noise	1	$\propto 1/f$	-3dB/oct -10dB/dec
Blue Noise	-1	$\propto f$	+3dB/oct +10dB/dec
Violet Noise	-2	$\propto f^2$	+6dB/oct +20dB/dec

according to the difference in the speed of the frequency decline, we generate a total of four types of colored noise, namely, red noise, pink noise, blue noise, and purple noise, the features of which are shown in Table III.

The colored noise is generated by the method in [31]. On the basis of Gaussian white noise, the noise is transferred to the frequency domain by Discrete Fourier Transform (DFT), the colored noise transform operation is performed on the noise in the frequency domain, and finally the colored noise time series is obtained by transferring to the time domain by Inverse Discrete Fourier Transform (IDFT). Firstly, the Gaussian white noise time domain expression $wgn[n]$ is generated, the frequency domain $WGN[k]$ is obtained by DFT transform, the colored noise $CN[k]$ in the frequency domain is obtained by operating on $WGN[k]$ in the frequency domain, and then the $CN[k]$ is subjected to IDFT to obtain the time domain colored noise sequence $cn[n]$.

$$WGN[k] = \frac{1}{N} \sum_{n=1}^N wgn[n] \cdot e^{-j2\pi \frac{nk}{N}} \quad (2)$$

$$CN[k] = WGN[k] \cdot k^{\frac{\alpha}{2}} \quad (3)$$

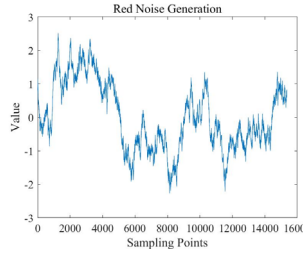


Fig. 8. Red noise genera-tion.

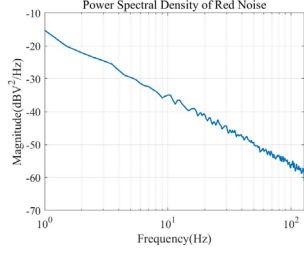


Fig. 9. Red noise power spectral density.

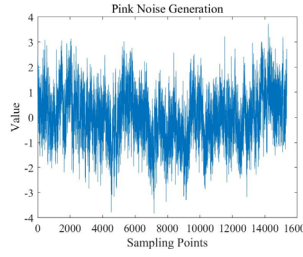


Fig. 10. Pink noise generation.

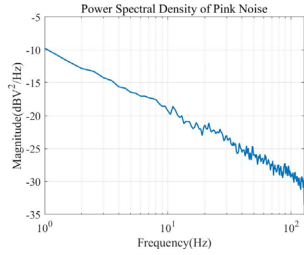


Fig. 11. Pink noise power spectral density.

$$cn[n] = \sum_{n=1}^N CN[k] \cdot e^{j2\pi \frac{nk}{N}} \quad (4)$$

In the experiment, we set the sampling frequency $fs = 256$, sampling time $T = 60$, which corresponds to the time of data sampling. The total number of samples for each type of noise is $N = 256 * 60 = 15,360$. Hanning window is used, and the window length is $winlen$ to select 2 multiple of sampling frequency, and periodic is selected for spectrum analysis. overlapping is $0.75 * winlen$, and the number of points of discrete Fourier transform $nfft$ is equal to $winlen$, and one-sided spectrum estimation is selected. α is taken as 2, 1, -1, -2 corresponding to 4 colored noises.

The results of the red noise generation are shown in **Fig. 8** and **Fig. 9**.

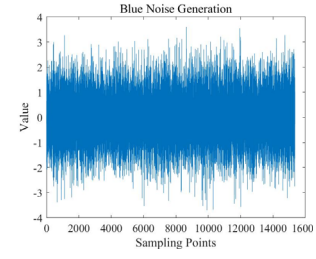


Fig. 12. Blue noise generation.

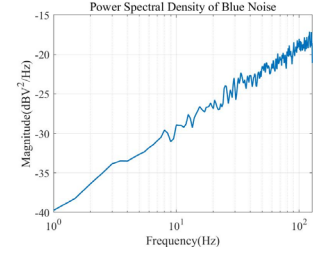


Fig. 13. Blue noise power spectral density.

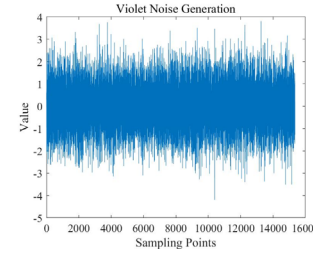


Fig. 14. Violet noise generation.

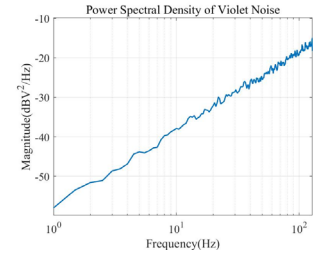


Fig. 15. Violet noise power spectral density.

The results of the pink noise generation are shown in **Fig. 10** and **Fig. 11**.

The results of the blue noise generation are shown in **Fig. 12** and **Fig. 13**.

The results of the violet noise generation are shown in **Fig. 14** and **Fig. 15**.

Noise Fusion Program:

Let there be a total of n data samples to be detected, and the botnet dataset has traffic features p , industrial control features q , and class labels k .

Then we can conclude that, $X = \{x_1, x_2, \dots, x_p\}$, $Y = \{y_1, y_2, \dots, y_q\}$, $Z = \{z_{1+p+q}, z_{2+p+q}, \dots, z_{k+p+q}\}$

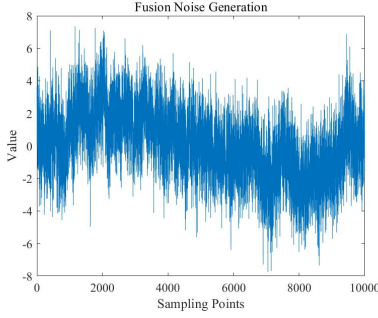


Fig. 16. Fusion noise generation.

Thereby the botnet dataset and class labels can be represented as follow, where $j = 1, \dots, n$:

$$\begin{aligned}
 S &= \{s_j \{X, Y, Z\}_j \\
 &= \{x_1, \dots, x_p; y_1, \dots, y_q; z_{1+p+q}, \dots, z_{k+p+q}\}_j \\
 &= \{x_{j1}, \dots, x_{jp}; y_{j1}, \dots, y_{jq}; z_{j(1+p+q)}, \dots, z_{j(k+p+q)}\}_j
 \end{aligned} \quad (5)$$

Let there are γ different noises, then the standardized noise set is denoted as $NOISE = \{noise_1, noise_2, \dots, noise_\gamma\}$, we define the noise fusion process as the symbol $\oplus$, and the fusion noise $noise_{all}$ can be denoted as follows:

$$noise_{all} = noise_1 \oplus noise_2 \oplus \dots \oplus noise_\gamma$$

Implanting noise into the dataset is:

$$\begin{aligned}
 S_{noise_{all}} &= \{C_N^F \cdot \rho \cdot noise_{all} \otimes S(z_{jM}) \\
 | j &= 1, \dots, n; N = p + q; M = 1 + p + q, \dots, k + p + q\}
 \end{aligned} \quad (6)$$

where C_N^F denotes that F features are selected as adversarial samples for generating the injection point among a total of N features of flow features and industrial control features, ρ denotes the noise disturbance coefficient with different multiplicity, and $\otimes$ denotes that noise is implanted into the dataset, then $S_{noise_{all}}$ is the dataset generated after adding noise.

In this experiment, we use five different noises, $noise_i | i \in (1, 5)$ corresponding to Gaussian noise, red noise, pink noise, blue noise, and purple noise, and mix the four colored noises with the same intensity, and select the first 10,000 noises to get the fusion noise.

The results of the fusion noise generation are shown in **Fig. 16**, **Fig. 17** and **Fig. 18**. The fusion noise preserves the inherent stochastic characteristics of noise while exhibiting structured temporal correlations, enabling it to effectively emulate realistic perturbations for adversarial sample generation.

The features that we collected are based on the flow's properties, and by altering the information in the uplink and downlink network traffic, we may produce adversarial samples for each. We only make minor changes to the flow features and do not change the industrial control features to ensure the correctness of the flow because botnet DoS traffic differs from man-in-the-middle attacks and injection attacks in that it does not change the values of the features.

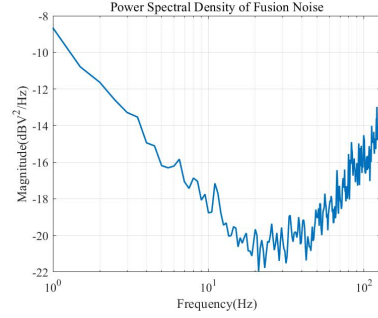


Fig. 17. Fusion noise power spectral density.

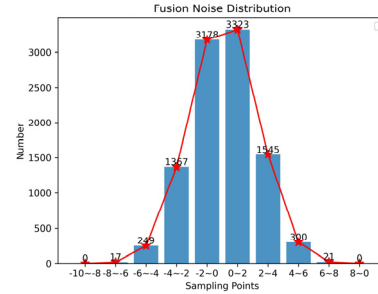


Fig. 18. Fusion noise distribution.

Once the detection models have been trained and evaluated, we generate adversarial samples to assess robustness. In our methodology, colored noise is treated as a time-series perturbation and is fused to alter the temporal characteristics of flows (i.e., packet timing and flow duration), whereas Gaussian noise is applied to the spatial characteristics of payloads (i.e., packet size and load statistics). We apply the same normalization and perturbation protocol across sample categories to ensure consistent interference. By separating temporal and spatial perturbation domains, the proposed scheme more faithfully reproduces realistic distortions in both transmission behavior and payload distribution, thereby improving the fidelity and representativeness of simulated adversarial attacks.

We leverage normalized noise and, based on varying perturbation coefficients, embed its absolute value into the normalized test set. From the 10,000 generated noises, a random subset, comprising 20%, is integrated into the test set. Given that the noise perturbation introduced is minimal, the overall mean and standard deviation of the samples remain largely unchanged. This subtlety allows an attacker, using the inverse process of normalization, to craft specific adversarial samples, further highlighting the necessity for models to be resilient against such potential intrusions.

IV. EXPERIMENT AND ANALYSIS

A. Experimental Setup and Methodology

In total, we trained 8 detection models including ML-DT [1], RF-IDS [1], Smote Tomek-Link XGBOOST (STL-BOOST) [4], IoT-Node ANN [7], Deep Residual 1DCN(DR-1DCN) [16], IoT botnet kill chain model (IoT BKCM) [4], PCA+MI [9] and PSO-SVM [5].

TABLE IV
CONFUSION MATRIX

Confusion Matrix	Predicted Class	
	TP	FN
True Class	FP	TN

For the machine learning model, we referred to the prototype proposed in the paper above to train the model. The neural network model was trained using the TensorFlow platform, IoT-Node ANN was trained for 100 epochs, DR-1DCN, PCA+MI, IoT BKCM for 30 epochs. The batch size of all four neural networks is chosen to be 512, the optimizer is chosen to be Adam. The training model preserves the best of all epochs.

For IoT-Node ANN containing 2 hidden layers and control the hidden layers to be 1/3 to 2/3 of the previous layer, 10 and 5 layers, respectively, and the activation function uses ReLU.

For DR-1DCN uses a one-dimensional deep convolutional neural network, each convolutional module consists of 5 layers, the first layer is a convolutional layer with 32 convolutional kernels and a convolutional kernel window of 5, the second layer is the activation function ReLU, the third layer is the same convolutional layer as the first layer, the fourth layer is the activation function ReLU, and the fifth layer is the maximal pooling layer, with a pooling scale of 5 and step size of 2. The model accepts the inputs and then passes through the 4 convolutional modules, then through the Flatten layer and the two Dense layers to get the output.

PCA+MI use a 2-layer LSTM, which is then trained with Flatten and Dense layers. The dimensions of the first and second layers of the LSTM are set to 32 and 16.

IoT BKCM incorporates a combination of 2 layers of 1D convolutional layers followed by 2 layers of LSTM layers. Subsequent processing is handled by Flatten and Dense layers. The output dimensions for the first and second convolutional layers are set to 128 and 64, respectively. The convolutional kernel window size is fixed at 5 with a stride (or step size) of 1.

The experiment is divided into two main parts:

First: training botnet detection models. All the models are trained for binary classification and multiple classification, respectively.

Second: The robustness of the detection model is evaluated using generated adversarial samples to observe the effect of different perturbation coefficient noises on the same model as well as the effect of different models on the same perturbation coefficient noise.

B. Performance Metric

TP: number of samples for which the classifier correctly predicted positive samples; FP: number of samples for which the classifier incorrectly predicted positive samples; TN: number of samples for which the classifier correctly predicted negative samples; FN: number of samples for which the classifier incorrectly predicted negative samples.

The confusion matrix is shown in Table IV.

(1) Accuracy, Which Represents the Proportion of Correctly Categorized Samples to the Total Sample, Calculated as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (7)$$

(2) Precision, Which Denotes the Proportion of Correctly Categorized Positive Samples to All Categorized Positive Samples, Calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

(3) Recall, Which Is the Proportion of Samples Predicted to Be Positive to All Positive Samples, Calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

(4) F1-Score, which is the comprehensive evaluation index of the model, combined with Accuracy and Recall comprehensive judgment, calculated as:

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (10)$$

C. Intrusion Detection Model Training

Each model is trained for both binary and multiclass classifications in accordance with our proposed feature extraction methodology. The dataset is partitioned with a ratio of 7:1:2 for training, validation, and testing, respectively. The confusion matrices depicted in the subsequent figures provide a clear visual representation of the efficacy of each model. Impressively, detection accuracies are observed to approach or even achieve 99% to 100%.

Such results underscore two significant achievements:

We have successfully devised an effective feature extraction technique specifically tailored for botnet identification within the industrial control system context.

The fusion noise perturbation model we employed has been meticulously and robustly trained.

However, one interesting observation emerges during the multiclass classification task without interference. Owing to the significant similarities between the two traffic types, all models occasionally misclassify a handful of samples from the “TCP Garbage Attack” category, marking them instead as belonging to the “Benign” category. This highlights the inherent challenges in distinguishing between traffic types that share closely related attributes.

IoT-Node ANN, DR-1DCN binary classification and multi-classification confusion matrices are shown in Fig. 19 and Fig. 20.

PCA+MI, IoT BKCM binary classification and multi-classification confusion matrices are shown in Fig. 21 and Fig. 22.

ML-DT, PSO-SVM binary classification and multi-classification confusion matrices are shown in Fig. 23 and Fig. 24.

RF-IDS, STL-BOOST binary classification and multiclassification confusion matrices are shown in Fig. 25 and Fig. 26.

20.

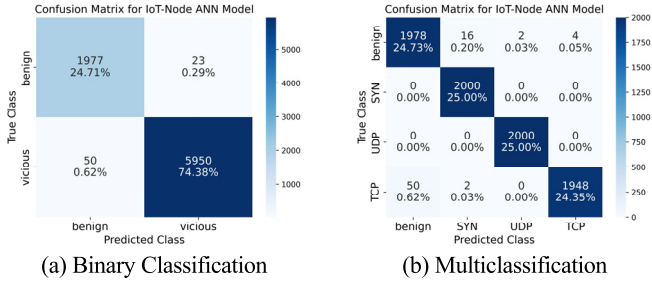


Fig. 19. IoT-Node ANN confusion matrix.

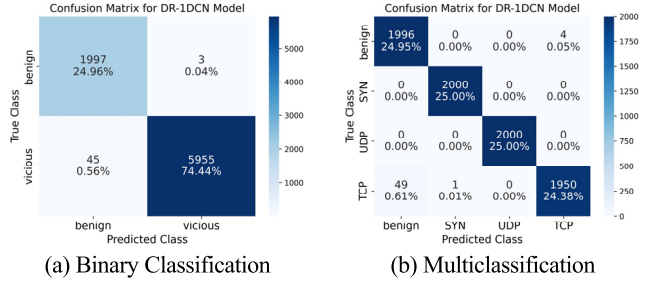


Fig. 20. DR-IDCN confusion matrix.

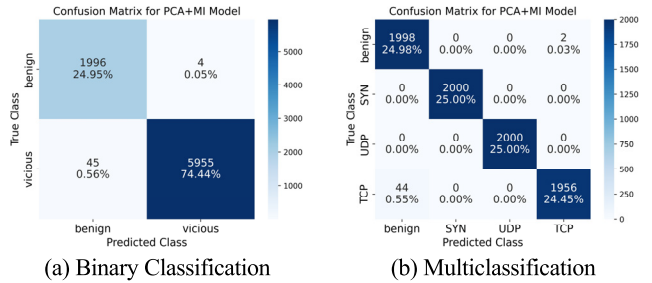


Fig. 21. PCA+MI confusion matrix.

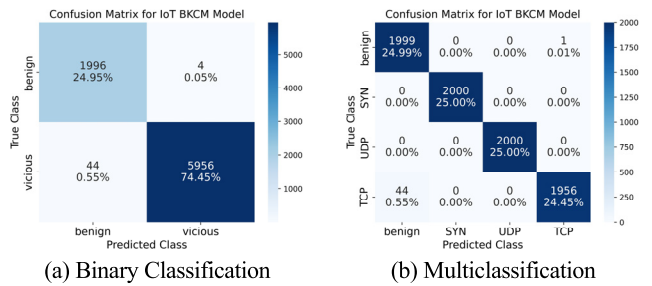


Fig. 22. IoT BKCM confusion matrix.

D. Adversarial Attack Evaluation

For comparison, a baseline random attack was implemented, which is commonly adopted in prior adversarial-traffic studies. This baseline perturbs the same subset of flow features as the proposed method but draws perturbations independently from a standard normal distribution and scales them to match the target perturbation coefficient. The identical normalization procedure and the same fraction of test samples are used as in the fusion-noise experiments, ensuring parity in perturbation magnitude and coverage. Importantly, like the proposed

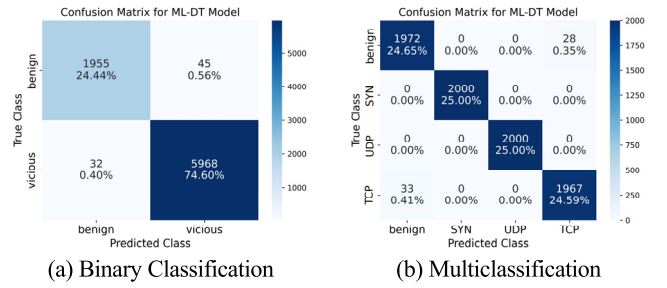


Fig. 23. ML-DT confusion matrix.

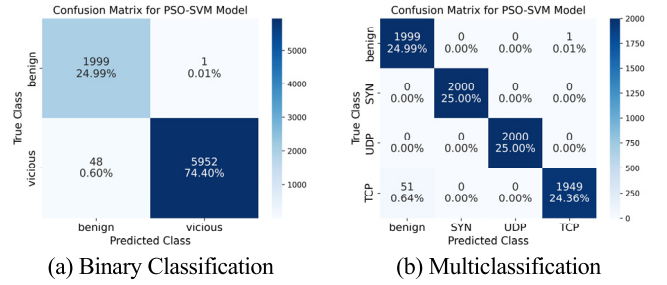


Fig. 24. PSO-SVM confusion matrix.

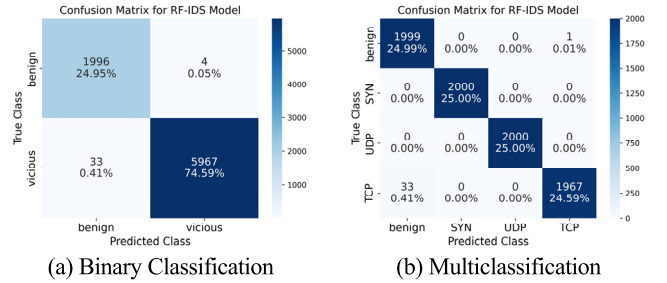


Fig. 25. RF-IDS confusion matrix.

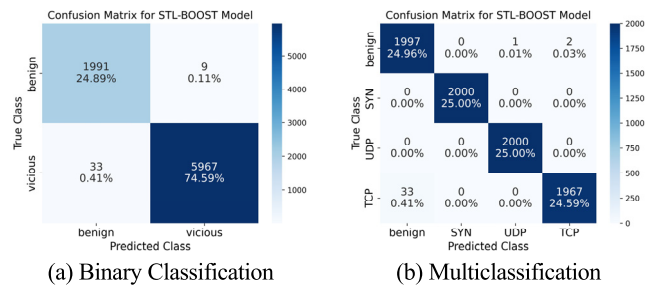


Fig. 26. STL-BOOST confusion matrix.

fusion-noise scheme, the random attack is executed under a fully black-box, gradient-free setting (no model queries or gradient information required). As such, it provides an appropriate and fair baseline for isolating the benefit of the fusion structure over unstructured perturbations.

For the purpose of to compare the detection abilities of eight different detection models on this adversarial sample, we created ten sets of adversarial samples with noise perturbation coefficients ranging from 0 to 0.5 spaced at 0.05. The blue line represents the detection model's accuracy, the yellow line

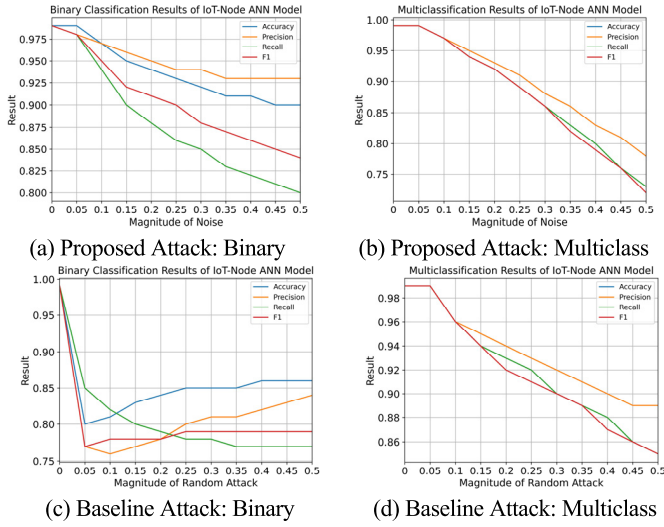


Fig. 27. IoT-Node ANN detection indicator curve.

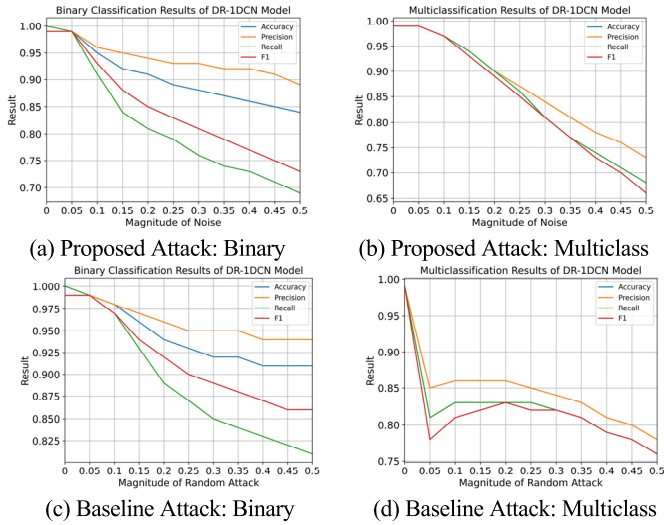


Fig. 28. DR-1DCN detection indicator curve.

represents its precision, the green line represents its recall, and the red line represents the F1-Score of the detection model.

The robustness of the models was evaluated based on their F1 score when subjected to adversarial samples with noise perturbations at 0.5 times the standard rate. It's evident that models exhibited greater vulnerability to adversarial samples in the multiclassification task as compared to the binary classification task.

This vulnerability is particularly pronounced under our structured fusion-noise attack. In the binary classification context, our method caused the performance metrics across all models to experience an average decline of 39%, with a maximum drop reaching 54%. This is a substantially more severe degradation than that achieved by the baseline random attack, which, as shown in Figures 27(c)–34(c), had a significantly milder impact on model performance. In the multiclassification scenario, the average decrease in metrics from our fusion-noise attack was around 44%, peaking at a substantial 57%. This effectiveness starkly contrasts with the baseline random

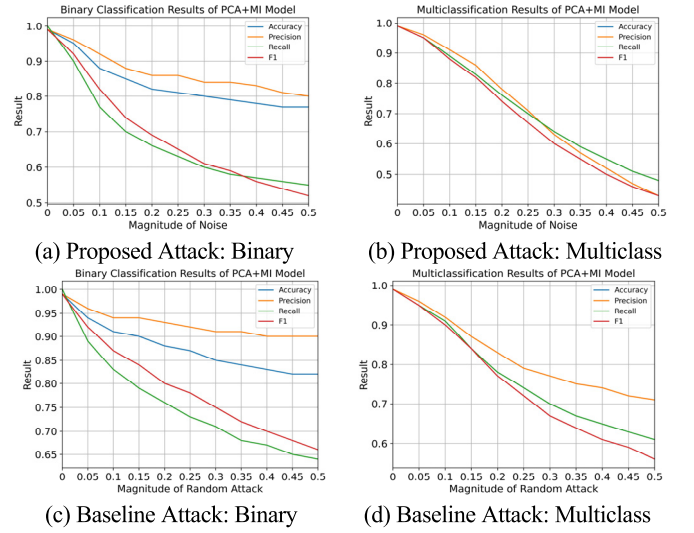


Fig. 29. PCA+MI detection indicator curve.

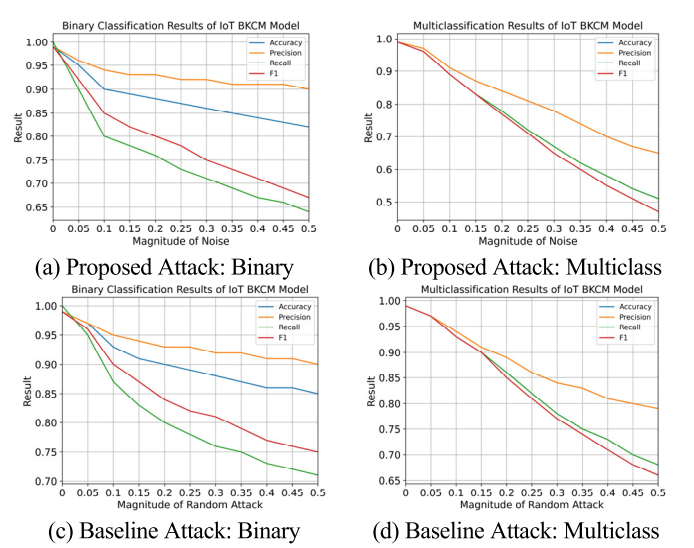


Fig. 30. IoT BKCM detection indicator curve.

attack's impact (shown in the (d) subplots of Figures 27–34), where the performance degradation was far less pronounced. This wide performance gap confirms that the structured, multi-domain nature of our fusion noise is key to its efficacy, rather than the mere addition of unstructured, random perturbations.

IoT-Node ANN is slightly better than DR-1DCN in adversarial sample tests for noise generation in binary and multiclassification tasks. The detection performance of IoT-Node ANN and DR-1DCN models with different perturbation coefficients is shown in Fig. 27 and Fig. 28.

IoT BKCM performs better than PCA+MI in both binary and multiclassification tasks, suggesting that the use of cross deep learning methods can somewhat increase the robustness of the detection model. The detection performance of PCA+MI and IoT BKCM models with different perturbation coefficients is shown in Fig. 29 and Fig. 30.

PSO-SVM is more noise-resistant than ML-DT, while PSO-SVM performs better than RF-IDS and STL-BOOST

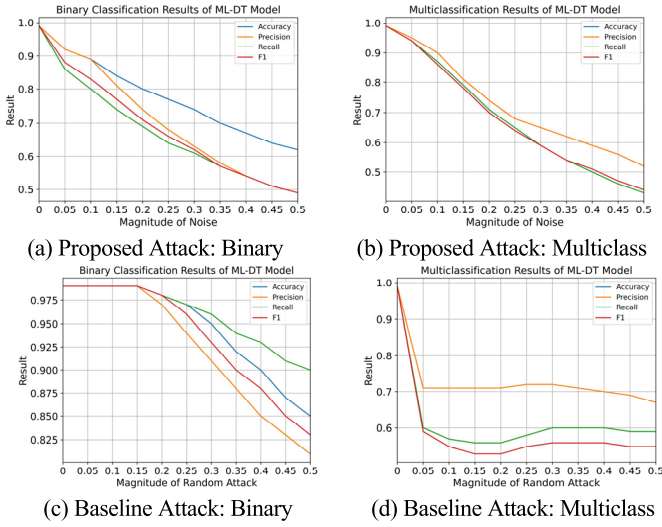


Fig. 31. ML-DT detection indicator curve.

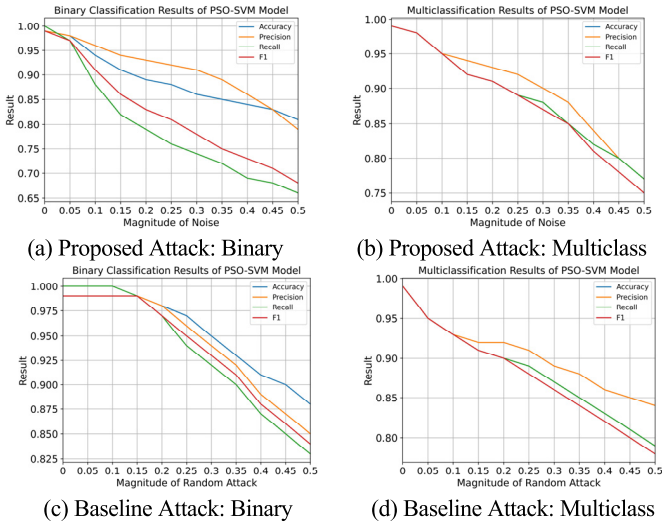


Fig. 32. PSO-SVM detection indicator curve.

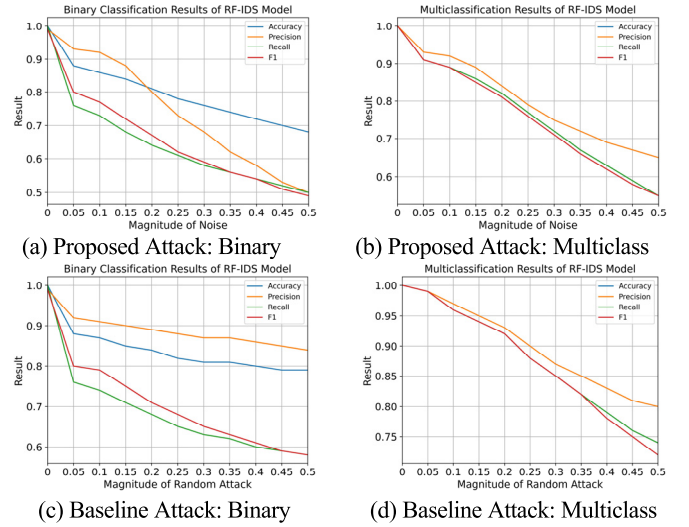


Fig. 33. RF-IDS detection indicator curve.

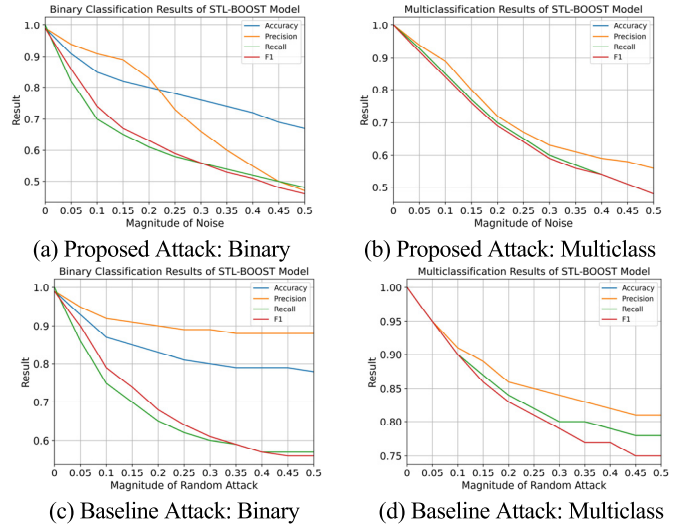


Fig. 34. STL-BOOST detection indicator curve.

algorithms, suggesting that tree-structured classifiers suffer from lower robustness when confronted with adversarial samples in the industrial control system environment. The detection performance of ML-DT and PSO-SVM models with different perturbation coefficients is shown in **Fig. 31** and **Fig. 32**.

It is obvious that the RF-IDS and STL-BOOST models, which are tree-structured models built utilizing an integrated approach of bagging and boosting, perform quite similarly in adversarial sample tests in the industrial control system environment. The detection performance of RF-IDS and STL-BOOST models with different perturbation coefficients is shown in **Fig. 33** and **Fig. 34**.

The reasons for the different performances of various models when faced with our proposed attack are as follows:

Certain models rely heavily on specific features of the input data, and if these features are perturbed by adversarial noise, the model's classification ability can significantly degrade. This phenomenon is particularly evident in models that have

not filtered out the effects of noise. Some models overfit the features of the training data and fail to generalize well to unseen adversarial examples which often exploit the vulnerabilities of these features.

This higher degree of performance degradation in multiclass classification tasks is mainly associated with the intrinsic complexity of their decision boundaries. In multiclass settings, the separation margins among classes in the feature space are typically narrower than those in binary classification, making the decision regions more sensitive to small perturbations. Consequently, even minor adversarial noise may shift samples across neighboring class boundaries, resulting in misclassification. Moreover, within ICS traffic data, several attack categories (such as TCP Garbage and UDP Flood) exhibit strong temporal and payload correlations with benign traffic. Such feature overlap increases the likelihood of confusion under noisy conditions. Additionally, neural networks trained with a softmax-based cross-entropy objective tend to distribute output probabilities among multiple competing classes,

which further amplifies uncertainty when adversarial perturbations are introduced. These factors collectively explain why multiclass detection models exhibit greater vulnerability to adversarial samples compared with their binary counterparts.

V. CONCLUSION

In the domain of industrial control system (ICS) botnet detection, this paper first introduces a novel feature extraction technique. Utilizing this method, we establish a botnet within the ICS environment and assess its effectiveness across four neural network models and four machine learning models. A dedicated ICS botnet detection dataset is also introduced. Empirical results indicate that all eight models proficiently identify ICS botnets utilizing the proposed feature extraction approach.

Additionally, this paper advances a black-box attack strategy, blending time-domain and frequency-domain sampling noise, to produce adversarial samples. We subsequently examine these samples against the eight models, comparing the effects of varying noise perturbation coefficients on model detection performance. Our research underscore the efficacy of the proposed adversarial sample generation strategy in reducing classification error rates. New adversarial sample attack methods can expose potential vulnerabilities in ICS botnet detection systems, prompting researchers and security experts to develop new defense mechanisms.

As a roadmap for future endeavors, we propose expanding the ICS intrusion detection dataset by:

Incorporating a broader spectrum of attacks across multiple protocols within the ICS, including in-depth analyses of the S7 and Modbus protocols. We also intend to refine the dataset by integrating man-in-the-middle, backdoor, injection, and other forms of attacks.

Evaluating and ranking feature significance to optimize feature selection. We will introduce perturbations selectively to highly significant features and contrast the implications of these perturbations across varied features on detection efficacy.

Existing models still make errors when distinguishing between types of traffic that have significant similarities and closely related attributes. Future work could focus on a more in-depth study of these types of traffic.

Generating effective adversarial samples requires substantial computational power and time. However, ICS typically demand high real-time performance. Attackers may lack sufficient time or computational resources to generate adversarial examples and promptly inject them into the real-time data streams of control systems. The attack overhead is not the focus of this study. In future work, we will explore the impact of the overhead associated with generating adversarial samples on the effectiveness of attacks.

To mitigate the real-time computational burden associated with fusion-noise generation, the noise synthesis process can be performed offline and stored as a reusable noise library. In this approach, frequency-domain transforms and noise-fusion procedures are executed a priori for a range of parameter settings (e.g., spectral exponent α , sampling window length, and perturbation coefficients ρ). During online evaluation or injection, precomputed noise segments are retrieved and

applied to normalized flow records with only lightweight operations (e.g., scaling, interpolation, or element-wise addition). This precomputation strategy substantially reduces runtime CPU/GPU requirements and enables rapid deployment in latency-sensitive ICS environments.

Future research should also investigate effective defense mechanisms to enhance the robustness of ICS botnet detection models against such hybrid adversarial perturbations. Potential approaches include adversarial training with fusion noise based samples, feature-space regularization to improve model generalization, and ensemble-based detection architectures that integrate temporal and spatial dependencies. Moreover, adaptive noise filtering and model uncertainty estimation could be explored to dynamically identify adversarial traffic in real-time ICS environments. These strategies can form a foundation for building resilient and self-adaptive detection frameworks capable of maintaining high reliability under adversarial conditions.

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